HENG ZENG XU (JASON)

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EDUCATION

Bachelor of Computer Science – Monash University Malaysia — Graduating Nov 2026 (coursework completed)
Coursework: Data Structures, Algorithms, Database Systems, Operating Systems, Parallel Computing, Mobile App Development

EXPERIENCE

Web Developer Intern – Impreszions Solutions — Dec 2025 – Feb 2026

- Built a museum-based educational treasure hunt (mobile web app) from scratch in React + Vite within an agile team, using the Google Maps JavaScript API for the game map, and deployed it on Firebase Hosting.
- Implemented proximity detection with the browser Geolocation API and Turf.js, then moved the distance check into Firebase Cloud Functions so the server, not the browser, decides when a player has reached a location.
- Built and deployed the production website for law firm Law Kuan Yew & Co. (lkyco.com.my), including responsive layouts and custom landing pages for the firm's services.
- Tested every layout at each breakpoint on iPhones, Android phones, tablets, laptops, and desktop monitors, fixing device-specific bugs before launch.

PROJECTS

Precisely – ERP & WhatsApp Ordering Agent for Chemical Distributors — 2026
Next.js · TypeScript · Supabase (Postgres + RLS) · Anthropic API · WhatsApp Cloud API | github.com/xengzuh/precisely

- Designed and built the whole system solo (22 tables, 31 routes): the UI and AI agents call one typed action registry, so all 25 business operations share a policy gate, an audit log, and undo.
- Built a WhatsApp ordering agent (Meta Cloud API + Claude tool loop) that reads live stock and records draft orders in ~10 s; a reply policy lets it answer 3 of 8 inquiry types alone, routing pricing, delivery, and safety questions to staff.
- Kept quantity math out of the LLM: a deterministic unit-conversion engine handles kg ↔ L by density and drums to base units, erroring on a missing density instead of guessing; 135 unit tests cover it and both agent policy gates.
- Replaced read-modify-write stock updates, which let two concurrent sales oversell, with atomic Postgres functions; closed a cross-tenant hole in 5 RPCs reachable from webhook-triggered agent runs, which bypass RLS via the service role.

CloudBoard – Real-Time Digital Signage Platform (Final Year Project) — 2026
Next.js · React · Supabase · TensorFlow.js · Google Cloud Vision · Video Intelligence API

- Built a web app to replace looping videos on TV displays: users upload content and edit each display's layout in a custom drag-and-drop canvas editor, and changes sync to the screens in real time through Supabase.
- Gated every upload behind automated moderation, with images checked by Google Cloud Vision SafeSearch and videos by the Video Intelligence API's explicit-content detection, so flagged media never goes live.
- Added audience-triggered ads: a webcam on each display feeds the mini PC behind it, which runs a pre-trained TensorFlow.js model on-device to count people and switches the screen to an ad when the count hits a set threshold.

Santorini – Desktop Strategy Board Game — 2025
Python · Tkinter · Pillow · PyInstaller | github.com/xengzuh/Santorini

- Built the Santorini board game as a desktop app with full rule enforcement (win and stalemate detection), three god powers, a per-player chess clock, and legal-move highlighting, shipped as a standalone Windows .exe.
- Split rules, god powers, and UI into independent layers: each god extends an abstract GodCard class built by a factory (a new god is one subclass), and the UI sits behind one facade; all rendering and animation is hand-drawn on the Tk canvas.

TECHNICAL SKILLS

Languages:	TypeScript, JavaScript, Python, Java, SQL, HTML/CSS
Backend & Data:	Node.js, PostgreSQL, Supabase (RLS, RPC functions), REST APIs, webhooks, schema design
Frontend:	React, Next.js (App Router), Vite, Tailwind CSS, WordPress
Cloud & APIs:	Firebase (Cloud Functions, Hosting), Vercel, Google Cloud Vision & Video Intelligence, Google Maps JavaScript API, WhatsApp Cloud API
AI & Tooling:	Anthropic API (tool use, structured outputs, prompt caching), TensorFlow.js, Vitest, Git